$$2x^2(2x+3) = 0.$$

Therefore, $x = 0, -\frac{3}{2}$ but $0, -\frac{3}{2} \notin (0,1)$.

Thus, $f'(x) > 0, \ \forall x \in (0,1)$.

Hence by the Rolle's theorem there do not exist $a, b \in (0,1)$ such that, $f(a) = 0 = f(b)$. Therefore the equation $f(x) = 0$ cannot have two roots in the interval $(0,1)$. But, $f(0) = -2 < 0$ and $f(1) = 1 > 0$ tells us the curve $y = f(x)$ crosses the x-axis between 0 and 1 only once by the Intermediate value theorem. Therefore the equation $x^4 + 2x^3 - 2 = 0$ has only one real root in the interval $(0,1)$. ■

As an application of the Rolle's theorem we have the following,

Example 7.23

Prove using the Rolle's theorem that between any two distinct real zeros of the polynomial

$$a_n x^n + a_{n-1}x^{n-1} + \cdots + a_1 x + a_0$$

there is a zero of the polynomial

$$na_n x^{n-1} + (n-1)a_{n-1}x^{n-2} + \cdots + a_1.$$

Solution

Let $P(x) = a_n x^n + a_{n-1}x^{n-2} + \cdots + a_1 x + a_0$. Let $\alpha < \beta$ be two real zeros of $P(x)$. Therefore, $P(\alpha) = P(\beta) = 0$. Since $P(x)$ is continuous in $[\alpha, \beta]$ and differentiable in (α, β) by an application of Rolle's theorem there exists $\gamma \in (\alpha, \beta)$ such that $P'(\gamma) = 0$. Since,

$$P'(x) = na_n x^{n-1} + (n-1)a_{n-1}x^{n-2} + \cdots + a_1$$

which completes the proof. ■

Example 7.24

Prove that there is a zero of the polynomial, $2x^3 - 9x^2 - 11x + 12$ in the interval $(2,7)$ given that 2 and 7 are the zeros of the polynomial $x^4 - 6x^3 - 11x^2 + 24x + 28$.

Solution

Applying the above example 7.23 with

$$P(x) = x^4 - 6x^3 - 11x^2 + 24x + 28, \ \alpha = 2, \ \beta = 7$$

and observing

$$\frac{P'(x)}{2} = 2x^3 - 9x^2 - 11x + 12 = Q(x), \text{ (say).}$$

This implies that there is a zero of the polynomial $Q(x)$ in the interval $(2,7)$. ■

For verification,

$$Q(2) = 16 - 36 - 22 + 12 = 28 - 58 = -30 < 0$$

$$Q(7) = 686 - 441 - 77 + 12 = 698 - 518 = 180 > 0$$

From this we may see that there is a zero of the polynomial $Q(x)$ in the interval $(2,7)$.

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